



6876 - An Stellar (JAGB/Carbon star) Distance to the Coma Cluster: Establishing Four New Determinations of the Hubble Constant

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(2) NGC-4911
	3		NIRCam Imaging	(5) NGC-4921
	2		NIRCam Imaging	(3) M-106

ABSTRACT

The Coma cluster, one of the richest nearby clusters of galaxies known, provides us with the unique opportunity to resolve the debate over the much-disputed value of the Hubble constant. And JWST is uniquely positioned to rapidly determine a precise and accurate value for the distance to the Coma using JAGB/carbon stars as standard candles. Knowing Coma's true distance immediately allows us to calculate four new measurements the Hubble constant, totally independent of the current Cepheid distance scale: (1) The recessional velocity of Coma simply divided by its distance gives a one-step value of H_0 at a distance that is on the threshold of the pure Hubble flow. (2) A distance to Coma will allow us to calibrate the absolute magnitudes of 12 historical Type Ia SNe in Coma which will, on their own, allow us to determine H_0 in the far field flow mapped by over 300 distant SNe Ia. (3) Combining the 12 supernovae in Coma with the 8 JAGB calibrated SNe in nearby galaxies will more than double the number of JAGB-based calibrators giving an even more robust estimate of H_0 . (4) A Coma distance will give the first stellar-based, zero-point calibration of the Fundamental Plane method which has over 4000 tracers of H_0 in the Hubble flow out to a redshift of 20,000 km/s.

Using ultra-wide filters (F150W2 & F322W2) on NIRCам it will be possible to complete this determination of H_0 with 2 (four-hour) exposures containing 5 Coma galaxies, a factor of 1.7 faster than standard wide filters. An absolute zero-point calibration (requiring less than 300 sec) will be made using N4258, which has a geometric distance. For a very modest amount of time the results will be of very high impact.

OBSERVING DESCRIPTION

This program has three pointings, two in the Coma cluster (one centered on NGC 4911 and another centered on NGC 4921) and the third centered on the photometric zero-point anchor galaxy NGC 4258. Using NIRCам, the two ultra-wide filters (F150W2 in the short-wavelength camera and F322W2 in the beam-split, long-wavelength camera) will allow us to detect and measure the carbon-star luminosity function in single visits to each of the galaxies.

The readout pattern will be DEEP8, with 6 GROUPS per INTEGRATION, 4 INTEGRATIONS per EXPOSURE, and a Total of 4 Dithers, for a total on-target exposure time of 18,681 sec.

Proposal 6876 - Targets - An Stellar (JAGB/Carbon star) Distance to the Coma Cluster: Establishing Four New Determinations of the ...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	NGC-4911	RA: 13 01 4.9103 (195.2704596d) Dec: +27 48 25.46 (27.80707d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Disk galaxies, Dwarf galaxies, Galaxy halos]				
	(3)	M-106	RA: 12 18 43.1367 (184.6797363d) Dec: +47 22 4.97 (47.36805d) Equinox: J2000	Epoch of Position: 2015.5	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Disk galaxies, Galaxy disks]				
	(5)	NGC-4921	RA: 13 01 26.1222 (195.3588425d) Dec: +27 53 9.63 (27.88601d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Galaxy disks, Galaxy halos]				

Proposal 6876 - Observation 1 - An Stellar (JAGB/Carbon star) Distance to the Coma Cluster: Establishing Four New Determinations ...

Observation	Proposal 6876, Observation 1										Tue Jan 27 21:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	NGC-4911	RA: 13 01 4.9103 (195.2704596d)			Epoch of Position: 2015.5					
			Dec: +27 48 25.46 (27.80707d)								
			Equinox: J2000								
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.									
Template		Category=Galaxy									
		Description=[Disk galaxies, Dwarf galaxies, Galaxy halos]									
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F322W2	MEDIUM8	10	4	16	4	16964.097		
Special Requirements	Aperture PA Range 117 to 137 Degrees (V3 116.94737309 to 136.94737309)										
	Fiducial Point Override NRCBS_FULL										

Proposal 6876 - Observation 3 - An Stellar (JAGB/Carbon star) Distance to the Coma Cluster: Establishing Four New Determinations ...

Observation	Proposal 6876, Observation 3										Tue Jan 27 21:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Observation 3) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(5)	NGC-4921	RA: 13 01 26.1222 (195.3588425d)				Epoch of Position: 2000				
			Dec: +27 53 9.63 (27.88601d)								
			Equinox: J2000								
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.								
Template	Category=Galaxy										
	Description=[Galaxy disks, Galaxy halos]										
Template	Module				Subarray				Target Placement		
	ALL				FULL				Module gap (large extended source)		
Dithers	#	Primary Dither Type			Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	INTRAMODULEBOX			4		SMALL-GRID-DITHER				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F322W2	DEEP8	6	4	16	4	18681.98		

Proposal 6876 - Observation 2 - An Stellar (JAGB/Carbon star) Distance to the Coma Cluster: Establishing Four New Determinations ...

Observation	Proposal 6876, Observation 2										Tue Jan 27 21:00:20 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	M-106	RA: 12 18 43.1367 (184.6797363d) Dec: +47 22 4.97 (47.36805d) Equinox: J2000			Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	Category=Galaxy Description=[Disk galaxies, Galaxy disks]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module A center (small extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULE		3		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W2	F322W2	BRIGHT2	5	2	6	3	676.417		
	2	F115W	F356W	SHALLOW4	5	1	3	3	773.047		
Special Requirements	Aperture PA Range 65 to 140 Degrees (V3 65.13434768 to 140.13434768) Fiducial Point Override NRCAS_FULL										